

A Level Mechanics

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Chapter 5: Moments and Equilibrium of Rigid Bodies

A Level Mechanics moves from particles to rigid bodies: forces no longer act “at a point” but at specific locations on an extended object, so the turning effect of a force (its *moment*) matters just as much as its size and direction. This chapter covers moments, the equilibrium of rigid bodies (rods, ladders, beams), and sliding and toppling — none of which arise when every object is treated as a single point, which is why this topic can feel unfamiliar even if you are already confident with forces on particles.

Why this chapter matters

The key conceptual shift in this chapter is moving from a *particle* model, where every force is drawn acting through one single point, to a *rigid body* model, where an object has size and shape and different forces can act at different, specific points along it. This introduces a completely new requirement: a rigid body can have zero resultant force (translational equilibrium, exactly as for a particle) and still rotate, so we must separately require the resultant *moment* to be zero (rotational equilibrium). The algebra involved is elementary — mostly multiplication, trigonometry, and solving linear equations — so the real difficulty is geometric reasoning: correctly identifying where a force acts, what line it acts along, and which point to take moments about. Because of this, diagram practice matters more here than in most other topics, and mark schemes reward a correctly labelled perpendicular distance and a well-chosen pivot point just as much as they reward the right final number.

5.1 Moments of a force

Concept

The **moment** of a force about a point measures its turning effect on a rigid body about that point. For a particle, only the resultant force matters. For a rigid body, this is no longer enough: a body can be in translational equilibrium (resultant force zero) and still rotate, so we also need the resultant moment to be zero.

The moment of a force F about a point O is defined as

$\text{force} \times \text{the perpendicular distance from } O \text{ to the line of action of } F$,

where the line of action is the straight line along which F acts, extended in both directions if necessary. Moments are measured in newton metres (N m).

A moment has a **sense**: clockwise or anticlockwise. A rigid body is in rotational equilibrium about a point when the sum of the clockwise moments about that point equals the sum of the anticlockwise moments about the same point.

Why the perpendicular distance, and not the distance to the force? A force only has a turning effect if its line of action misses the pivot. If a force's line of action passes exactly through the pivot, it has no leverage at all and produces zero moment, however large it is. The perpendicular (right-

angle) distance from the pivot to that line is what measures how much leverage the force has — this is why it is the perpendicular distance that appears in the formula, not the distance to the point where the force happens to be applied.

If a force does not act perpendicular to the object, you cannot read the perpendicular distance straight off the diagram — you must find it using trigonometry. This step is the single most common source of error in this topic, so always sketch the perpendicular distance as a separate line on your diagram before writing any equation.

Key vocabulary

Term	Meaning
Moment	The turning effect of a force about a point, equal to force $\times$ perpendicular distance from the point to the force's line of action
Line of action	The infinite straight line along which a force acts
Perpendicular distance	not the distance to the point of application, unless the force happens to act perpendicular to the object
Sense	Whether a moment tends to turn the body clockwise or anticlockwise about the chosen point
Resultant moment	The sum of all moments about a point, taking sense into account (clockwise and anticlockwise moments have opposite signs)

Key formulas

KEY FORMULA

$$M = F d$$

$$M = F r \sin\theta$$

Here, d is the perpendicular distance from the pivot to the force's line of action, r is the distance from the pivot to the actual point where F is applied, and θ is the angle between the force F and the line joining the pivot to that point. The two formulas are equivalent: $d = r \sin\theta$, so the second formula is simply a convenient way to compute the perpendicular distance when you know r and θ rather than d directly.

Worked example

EXAMPLE 5.1

Question. A light rod OB has length 1.5 m. A force of 20 N is applied at B , acting at 40° above the rod. Find the moment of the force about O , stating its sense.

Given: rod length $OB = 1.5$ m, force $F = 20$ N at B , angle between the force and the rod $= 40^\circ$.

Step 1 — find the perpendicular distance. Sketch the rod OB and the force at B , then drop a perpendicular line from O to the force's line of action. This perpendicular distance is the opposite

side of a right-angled triangle with hypotenuse $OB = 1.5 \text{ m}$ and included angle 40° , so it is given by $d = OB \times \sin 40^\circ$. **M1**

$$d = 1.5 \times \sin 40^\circ$$

$$d = 1.5 \times 0.6428... = 0.9642... \approx 0.964 \text{ m (3 s.f.)} \quad \textbf{A1}$$

Step 2 — multiply the force by this perpendicular distance. By the definition of a moment, $M = F \times d$. **M1**

$$M = 20 \times 0.9642... = 19.28... \approx 19.3 \text{ N m (3 s.f.)} \quad \textbf{A1}$$

Step 3 — determine the sense. Picture the force pulling B upward and away from the rod: this rotates the rod anticlockwise about O. **B1**

Answer: The moment of the force about O is 19.3 N m , anticlockwise.

WHY THIS TOPIC OFTEN FEELS HARD AT FIRST

In particle mechanics, every force is drawn acting through the single point representing the object, so the question “how far is the pivot from this force’s path?” never comes up — every force effectively acts at zero distance from every other force. Moments introduce that question for the first time, and it takes practice to see a force’s line of action as an infinite line rather than just an arrow starting at a point. A quick experiment helps a great deal: push on a real rod or ruler pivoted at one end, and ask yourself which line the force travels along, and how far the pivot is from that line — this makes the abstract idea concrete before any algebra is attempted. It is also worth flagging early that taking moments about a point that lies exactly *on* a force’s line of action deliberately makes that force’s moment zero — this is a useful technique (used throughout 5.2), not a mistake to avoid.

Common exam mistakes

- Using the distance to the point of application instead of the perpendicular distance to the line of action.
- Forgetting the $\sin\theta$ factor when the force is not perpendicular to the rod.
- Losing track of sense (clockwise vs anticlockwise) partway through a multi-force problem.
- Giving a moment with no units, or the wrong units (N instead of N m).

5.2 Equilibrium of a rigid body

Concept

A rigid body in equilibrium must satisfy **two independent conditions at the same time**, and A Level questions expect both to be used:

1. The resultant force is zero — resolve in two perpendicular directions, usually horizontal and vertical, exactly as you would for a particle.
2. The resultant moment about any point is zero.

A useful fact. If the resultant force on a rigid body is zero, then once the resultant moment about *one* point is zero, it is automatically zero about *every* point. This is why you are free to choose whichever point makes the moments equation simplest, rather than being told which point to use.

Choosing a pivot strategically. Because you may take moments about any point, the standard technique is: take moments about a point where an unknown force acts. That force's perpendicular distance from the chosen point is then zero, so its moment is zero — it drops out of the equation entirely, however large it is, leaving one equation in a single unknown. This is the single most useful problem-solving trick in this chapter, and it is worth naming explicitly every time you use it.

For a **uniform** rod or beam, the weight acts at the midpoint (the centre of mass), by symmetry. For a **non-uniform** rod, the centre of mass is not at the midpoint — its position will be given in the question (for example, “the centre of mass is 1.2 m from A”) and must be used in place of the midpoint.

Key vocabulary

Term	Meaning
Rigid body	An object with size and shape that does not bend, so forces can act at different, specific points on it
Centre of mass	The single point at which the whole weight of a body may be taken to act
Uniform rod/beam	A rod whose mass is evenly distributed, so its centre of mass is at its midpoint
Non-uniform rod/beam	A rod whose centre of mass is not at its midpoint; its position must be given or found
Reaction	The force exerted by a support, pivot, or surface on a body in contact with it
Equilibrium	A state in which the resultant force and the resultant moment on a body are both zero

Key formulas

KEY FORMULA

$$\Sigma F^x = 0, \quad \Sigma F^y = 0, \quad \Sigma M^O = 0$$

The first two equations are exactly the resolving equations used for particles. The third is new: it says the resultant moment about a chosen point O must also be zero.

Worked example

EXAMPLE 5.2

Question. A uniform ladder AB of length 4 m and weight 200 N rests with end A on rough horizontal ground and end B against a smooth vertical wall, making an angle of 60° with the ground. Find (a) the normal reaction from the wall, (b) the normal reaction from the ground, and (c) the friction force at A.

Given: ladder length AB = 4 m, weight W = 200 N (uniform, so acts at the midpoint, 2 m from A),

angle with ground = 60° , ground rough, wall smooth.

Step 1 — set up the forces. At A (rough ground): normal reaction R , acting vertically, and friction F , acting horizontally, since the ground is rough and can resist sliding. At B (smooth wall): only a normal reaction S , acting horizontally, since a smooth surface provides no friction. The weight, 200 N , acts vertically downward at the midpoint of the ladder, 2 m from A. Sketch all four forces on the ladder before continuing. **B1**

Step 2 — resolve vertically. The wall is smooth, so it contributes no vertical force at all; the weight is the only other vertical force. **M1**

$$R = 200\text{ N} \quad \text{A1}$$

Step 3 — take moments about A. Taking moments about A is the strategic choice here: both unknown forces at A (R and F) act through A itself, so their perpendicular distance from A is zero and they both drop out of the moments equation, leaving a single equation in the one remaining unknown, S . The perpendicular distance from A to the line of action of S (horizontal, at B) is the vertical height of B, which is $4 \sin 60^\circ$. The perpendicular distance from A to the line of action of the weight (vertical, through the midpoint) is the horizontal distance to the midpoint, which is $2 \cos 60^\circ$. **M1**

$$S \times (4 \sin 60^\circ) = 200 \times (2 \cos 60^\circ) \quad \text{A1}$$

Now substitute $\sin 60^\circ = 0.8660\dots$ and $\cos 60^\circ = 0.5$, then rearrange to make S the subject:

$$S \times 3.464\dots = 200 \times 1$$

$$S = 200 / 3.464\dots = 57.73\dots \approx 57.7\text{ N} \text{ (3 s.f.)} \quad \text{A1}$$

Step 4 — resolve horizontally. The only horizontal forces on the ladder are S (from the wall) and F (friction at A), so for equilibrium these must be equal and opposite in effect. **M1**

$$F = S = 57.7\text{ N} \quad \text{A1}$$

Answer: $R = 200\text{ N}$, $S = 57.7\text{ N}$, $F = 57.7\text{ N}$. The ladder is in equilibrium provided the ground is able to supply this much friction force.

WHY THIS PROBLEM IS A GOOD TEST OF THE WHOLE IDEA

This is the chapter's signature “double equilibrium” question, because it forces you to combine two skills that are usually practised separately: resolving forces (familiar from particle mechanics) and taking moments about a strategically chosen point (new to this chapter). It is easy to underestimate how much this combination adds to a problem. The fix is procedural: explicitly name the technique — “take moments about a point where an unknown force acts, to eliminate it” — every time you use it, and always draw the full force diagram first, labelling every perpendicular distance on it, before writing a single equation. Most errors in this style of question come from trying to write the moments equation directly from the words of the problem, without a fully labelled diagram to work from.

Common exam mistakes

- Taking moments about a point that does not eliminate an unknown, creating an equation with two unknowns unnecessarily.

- Using the ladder's full length instead of the correct perpendicular distance (forgetting the $\sin\theta$ or $\cos\theta$ factor) when taking moments.
- Placing the weight of a uniform rod at the wrong point (not the midpoint), or using the midpoint for a non-uniform rod.
- Sign errors from mixing clockwise and anticlockwise moments on the same side of the equation.
- Forgetting that a smooth surface contributes only a normal reaction, with no frictional component.

5.3 Sliding and toppling

Concept

A rigid body resting on a rough surface, subject to an increasing applied force, can lose equilibrium in one of two distinct ways, and A Level questions often ask you to work out which happens first.

Sliding: the applied force overcomes the maximum available friction. The body is on the point of sliding when friction is limiting, meaning $F = \mu R$ (μ is the coefficient of friction, R is the normal reaction).

Toppling: the applied force creates enough moment to rotate the body about an edge of its base. The body is on the point of toppling when the normal reaction R acts exactly at that edge — as the applied force increases, the normal reaction shifts toward the edge in the direction of tipping, and toppling begins the instant it reaches that edge. This lets you find the toppling condition directly by taking moments about the edge.

To decide which failure mode happens first as an applied force increases from zero, find the value of the force required for sliding and the value required for toppling separately, then compare them: **the smaller of the two values is reached first**, and that determines the mode of failure. Toppling problems use the same “take moments about a well-chosen point” method as 5.2 — here, the chosen point is the edge about which toppling would occur, because the vanishing normal reaction has zero moment about that exact point.

Key vocabulary

Term	Meaning
Limiting equilibrium	The state in which friction is at its maximum possible value, $F = \mu R$
Coefficient of friction (μ)	The ratio of maximum friction to normal reaction at a rough contact
On the point of sliding	The body is about to slide: friction is limiting ($F = \mu R$) but the body has not yet moved
On the point of toppling	The body is about to rotate about an edge: the normal reaction acts exactly at that edge
Line of action of the weight	The vertical line through the centre of mass; toppling begins once this line passes outside the base

Key formulas

KEY FORMULA

$$F = \mu R$$

$$\Sigma M^{\text{edge}} = 0$$

Worked example

EXAMPLE 5.3

Question. A uniform rectangular block of weight 80 N stands on rough horizontal ground, with width 0.6 m (so the centre of the base is 0.3 m from each vertical edge) and height 1.6 m. The coefficient of friction between the block and the ground is 0.5. A horizontal force P is applied at height 0.8 m (mid-height) on a vertical face and gradually increased from zero. Determine whether the block slides or topples first, and find the value of P at which this occurs.

Given: weight $W = 80$ N, half-width to nearer edge = 0.3 m, height of application of $P = 0.8$ m, $\mu = 0.5$, P horizontal.

Step 1 — find the sliding condition. The block is on the point of sliding when friction is limiting. Because P is horizontal, it has no vertical component, so the normal reaction R simply equals the weight. **M1**

$$R = 80 \text{ N}$$

$$P_{\text{slide}} = \mu R = 0.5 \times 80 = 40 \text{ N} \quad \text{A1}$$

Step 2 — find the toppling condition. At the point of toppling, the normal reaction acts exactly at the leading bottom edge (the edge the block is about to tip over). Taking moments about that edge is the strategic choice: it makes R 's perpendicular distance zero, eliminating the unknown reaction from the equation entirely. The moment of P about that edge is $P \times 0.8$ (its height above the ground); the moment of the weight about that edge is 80×0.3 (its horizontal distance from the edge, since the weight acts through the centre of the base). **M1**

$$P_{\text{topple}} \times 0.8 = 80 \times 0.3 \quad \text{A1}$$

$$P_{\text{topple}} = (80 \times 0.3) / 0.8 = 24 / 0.8 = 30 \text{ N} \quad \text{A1}$$

Step 3 — compare the two conditions. $P_{\text{topple}} = 30$ N is smaller than $P_{\text{slide}} = 40$ N, so as P increases from zero, the toppling condition is reached first. **B1**

Answer: The block topples before it slides, when $P = 30$ N.

WHY TOPPLING IS THE HARDER IDEA TO GRASP

Sliding is a natural extension of friction problems already met with particles, since $F = \mu R$ is unchanged. Toppling is the genuinely new idea here: it requires picturing the base of the object as having a definite width, and recognising that the normal reaction's position along that base can shift as the applied force P increases — it is not fixed at the centre. A simple experiment makes this vivid: push a box low down on a table and it slides; push the same box high up and it tips instead. This shows directly why both the height at which the force is applied and the half-width of the base determine the outcome, and why the question “which happens first?” can never be answered by intuition alone — both conditions must be calculated and compared.

Common exam mistakes

- Comparing P_{slide} and P_{topple} without actually computing both, or assuming sliding always happens first because it is the more familiar idea.
- Using the full width of the base instead of the distance from the centre of mass to the edge when taking moments for toppling.
- Forgetting that R equals the weight only when the applied force is horizontal; a non-horizontal applied force changes R and must be resolved first.
- Taking moments about the wrong edge (the trailing edge instead of the leading edge in the direction of tipping).

Chapter summary

Idea	Key point
Moment	$M = F \times \text{perpendicular distance from pivot to line of action}$, in N m, with a clockwise or anticlockwise sense
Rigid body equilibrium	Requires both resultant force zero (resolve in two directions) and resultant moment zero (take moments about a chosen point)
Strategic pivot choice	Taking moments about a point where an unknown force acts eliminates that force from the equation
Uniform vs non-uniform rod	Weight acts at the midpoint for a uniform rod, but at a stated (different) point for a non-uniform rod
On the point of sliding	Friction is limiting: $F = \mu R$
On the point of toppling	The normal reaction acts exactly at the edge of the base; take moments about that edge
Sliding vs toppling	Compute both P_{slide} and P_{topple} and compare — the smaller one occurs first

Common exam mistakes

- Using the distance to the point of application rather than the true perpendicular distance to the line of action.
- Choosing a pivot point that leaves two unknowns in the moments equation instead of one.
- Placing a uniform rod's weight anywhere other than its midpoint.
- Mixing clockwise and anticlockwise moments on the same side of an equation (sign errors).
- Assuming sliding always happens before toppling without comparing both conditions explicitly.

You should be able to...

- Calculate the moment of a force about a point, including when the force is not perpendicular to the body, and state its sense.

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- Set up and solve equilibrium problems for a rigid body (rod, beam, or ladder) by resolving forces in two directions and taking moments about a chosen point.
 - Choose a pivot point strategically to eliminate an unknown force from a moments equation.
 - Distinguish uniform from non-uniform rods and locate the centre of mass correctly in each case.
 - Derive and apply the conditions for a body to be on the point of sliding ($F = \mu R$) and on the point of toppling (R acts at the edge of the base), and determine which occurs first.